

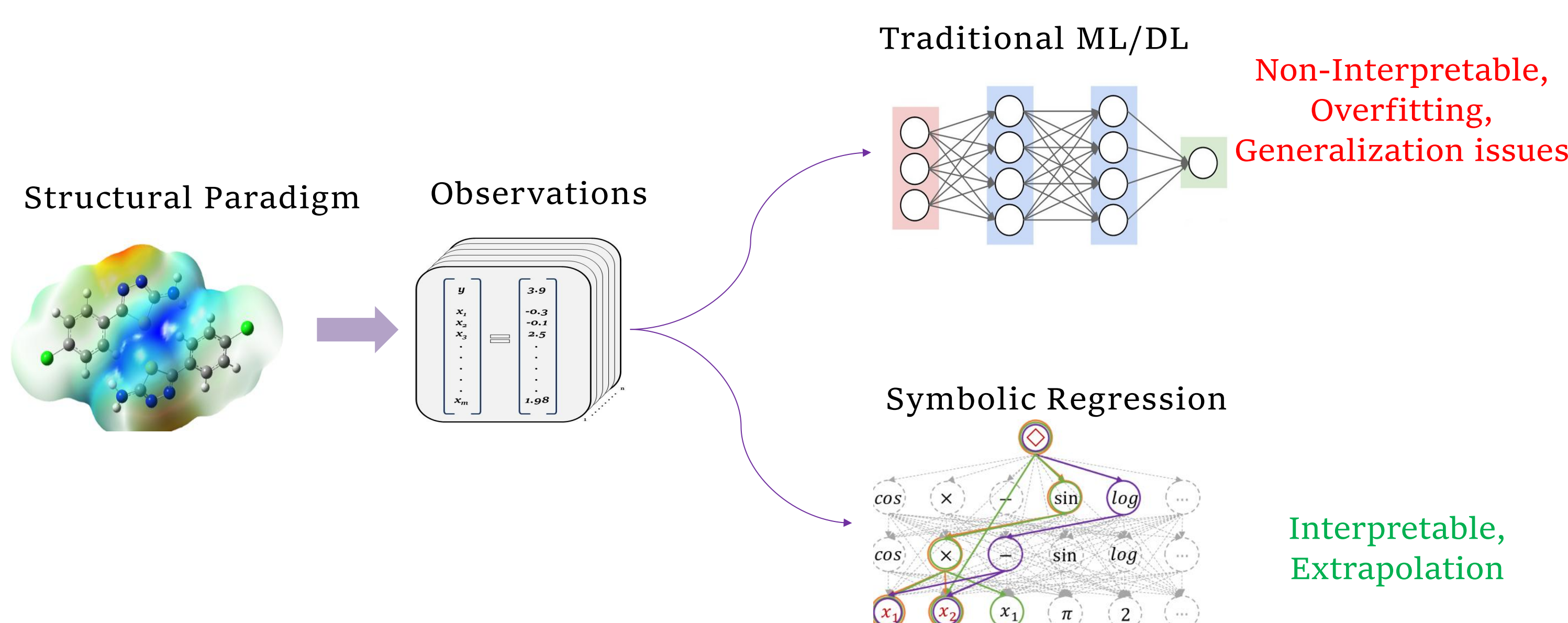


SyMANTIC: An Efficient Symbolic Regression Method for Interpretable and Parsimonious Model Discovery in Science and Beyond

Madhav Muthyala and Joel A. Paulson

Motivation

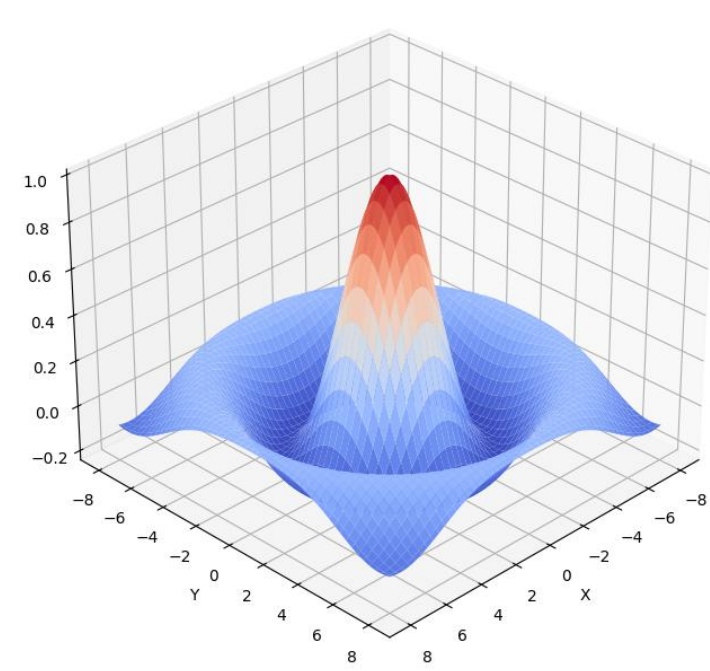
- In fields where decisions are based on the model predictions, understanding the “*why*” behind those predictions can lead to more effective interventions and a deeper understanding of the system’s dynamics.



Symbolic Regression

Symbolic Regression

- The objective is to search equation form $f(\mathbf{x}; \theta)$ that best fits the data by selecting appropriate mathematical operators, variables and constants.



$$f(x, y) = \sin(x^2 + y^2) / \sqrt{x^2 + y^2}$$

Existing Approaches

Genetic Programming (GP)

Evolves the symbolic expressions search space using operations like selection, cross-over and mutation.

✓ Searches the space with minimal prior assumptions

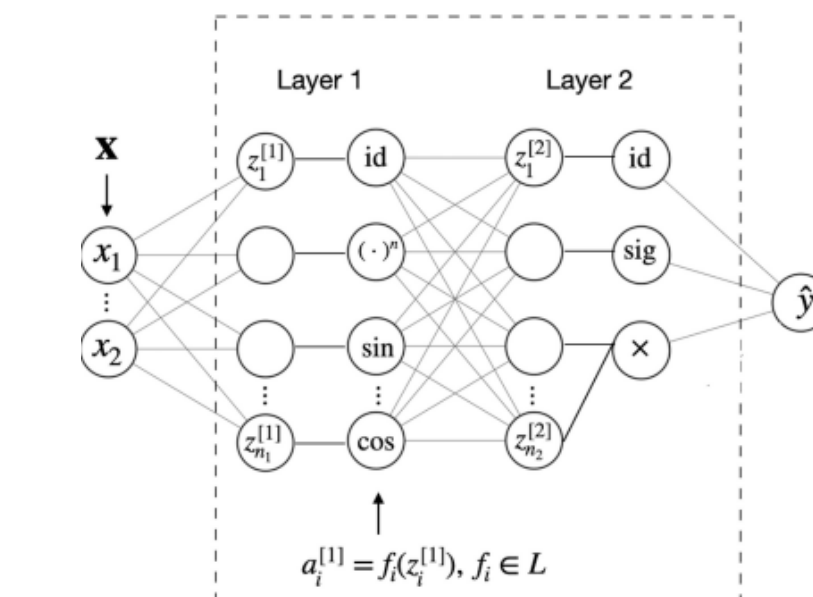
✗ Expensive, Complex models

Deep Learning Approaches

Leverages neural networks with operators as activation functions to generate the symbolic expressions with gradient-based optimization

✓ Flexibility and Scalability in handling high dimensional datasets

✗ Large data requirement, overfitting, and lacks intermediate interpretability



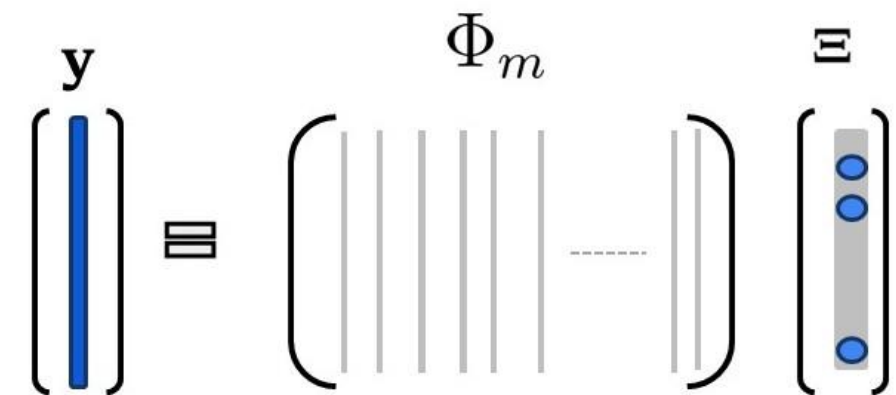
Some of the popular methods are Gplearn, PySR, Eureqa

Sparse Regression Approaches

Uses regularization penalties to promote the sparsity in the model coefficients associated with the curated expanded expressions space

✓ Efficient, scalable and easy to implement

✗ Careful tuning of hyperparameters



Some of the popular methods are SINDy, LASSO

SyMANTIC

- SyMANTIC** (Symbolic Modeling with Adaptive iNtelligent feaTure exPansion and Compressed sensing) is a combination strategy of iterative feature expansion and smart sparse regression to build the pareto of interpretable models.

- For a given data ($\mathbf{z} \in \mathbb{R}^d$) and operator set ($\mathcal{O}$) SyMANTIC executes these steps iteratively:

- Builds the interaction feature space according to

$$\phi_l(\mathbf{x}) = \{\mathcal{O}[z_i(\mathbf{x}), z_j(\mathbf{x})], \forall z_i(\mathbf{x}), z_j(\mathbf{x}) \in \phi_{l-1}(\mathbf{x})\}$$

- Calculates the complexity of the expressions resulted using

$$C(f) = K(f) \log_2 B(f)$$

- Builds the pareto set of interpretable models by running the **C²-SISSO** algorithm:

$$(\mathcal{M}_{c,l}, \mathcal{L}_{c,l}, \mathcal{C}_{c,l}) = \text{C}^2\text{-SISSO}(\mathbf{y}, \Phi_l, \lambda_c), \quad \forall (c, l) \in \{1, \dots, n_{\text{comp}}\} \times \{1, \dots, n_{\text{exp}}\}$$

for λ_c in complexity filter:

- Based on the complexity filter (λ_c) screens the space of expressions using for sparse regression

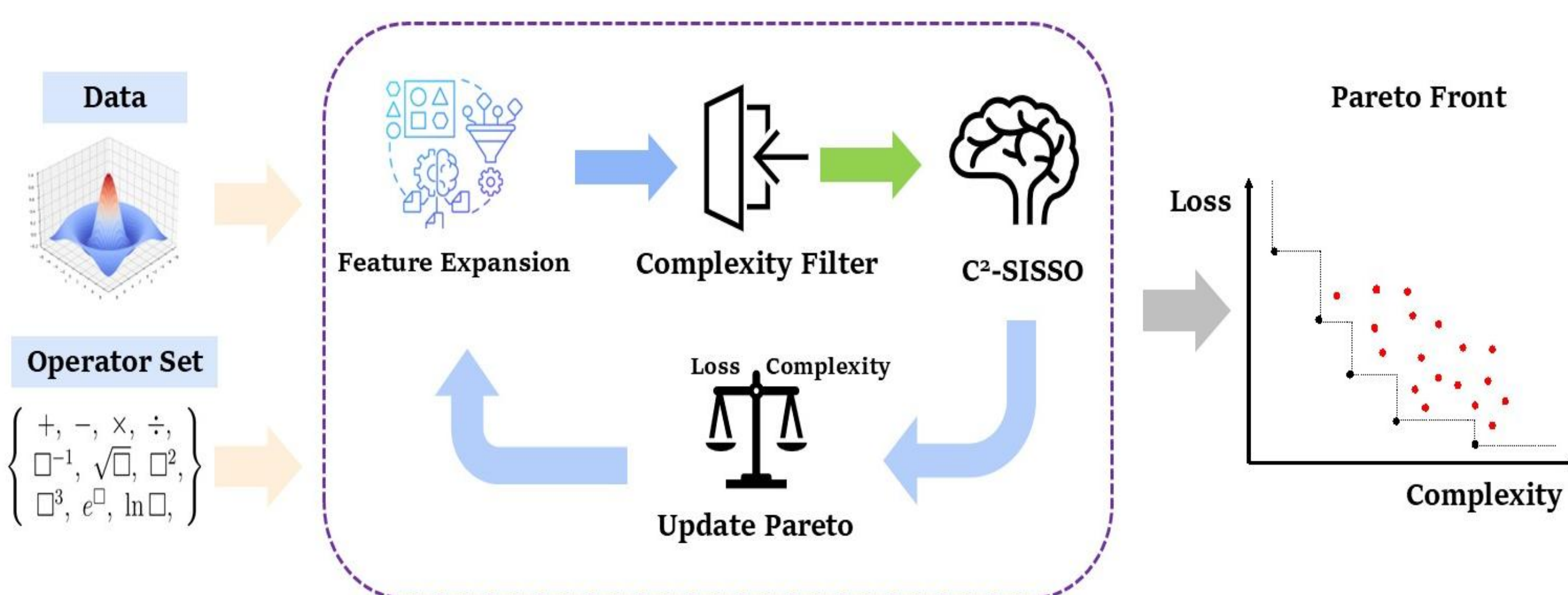
$$\omega = (\omega_1, \dots, \omega_d) = \Phi^\top \mathbf{y}$$

- Builds the iterative residual linear models using L_0 regularization (Number of non zero terms)

$$\mathbf{r}_t = \mathbf{y} - \Phi_{S_t} \mathbf{E}_t \mathbf{c}_t \quad \text{where} \quad \mathbf{c}_t = (\mathbf{E}_t^\top \Phi_{S_t}^\top \Phi_{S_t} \mathbf{E}_t)^\top \mathbf{E}_t^\top \Phi_{S_t}^\top \mathbf{y}$$

- Update the pareto set of models

$$\mathcal{P}_i = \text{Update_Pareto}(\mathcal{P}_{i-1}, \mathcal{M}_{c(i),l(i)}, \mathcal{L}_{c(i),l(i)}, \mathcal{C}_{c(i),l(i)}), \quad \forall i \in \{1, \dots, n_{\text{exp}} n_{\text{comp}}\}$$

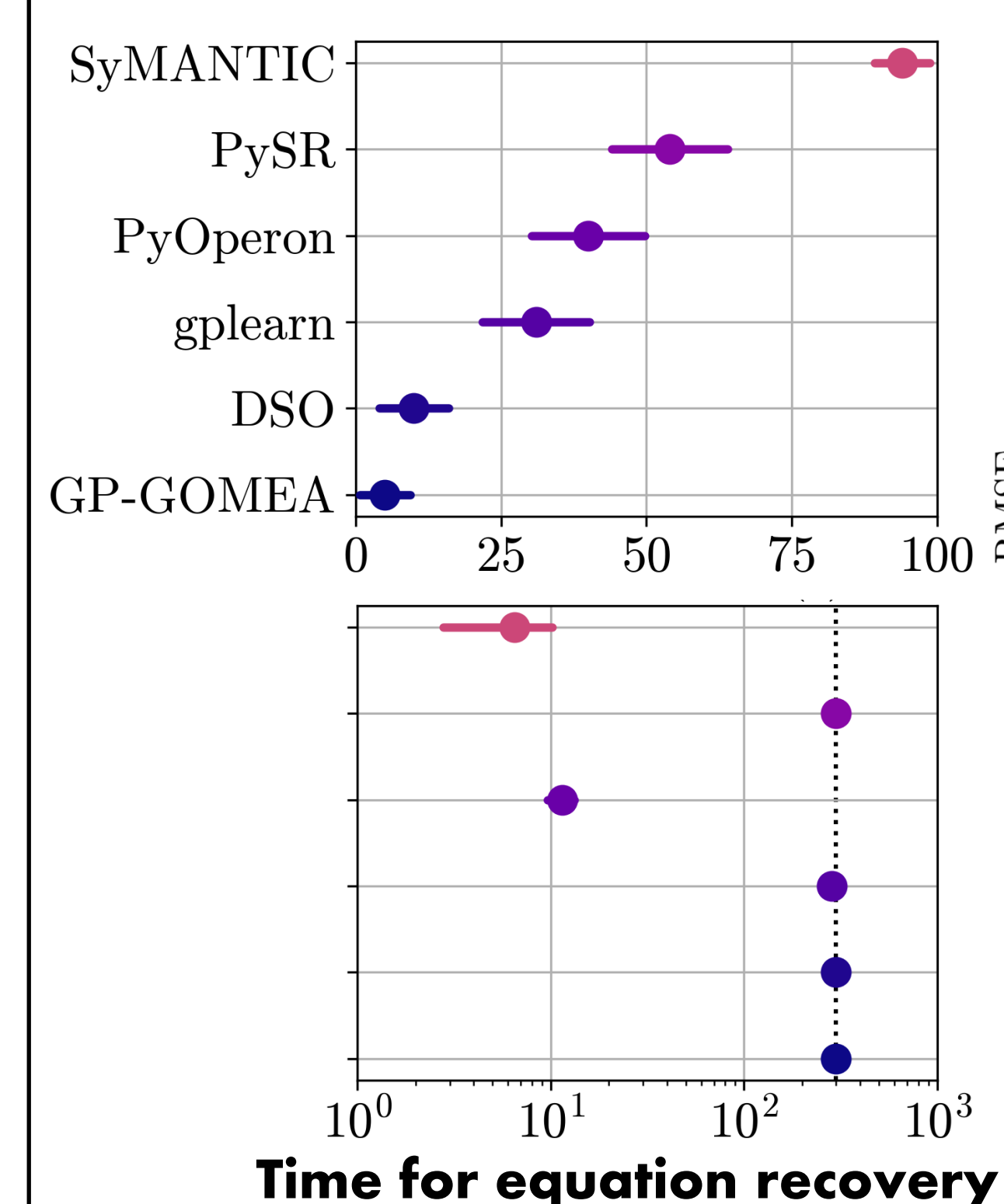


Results

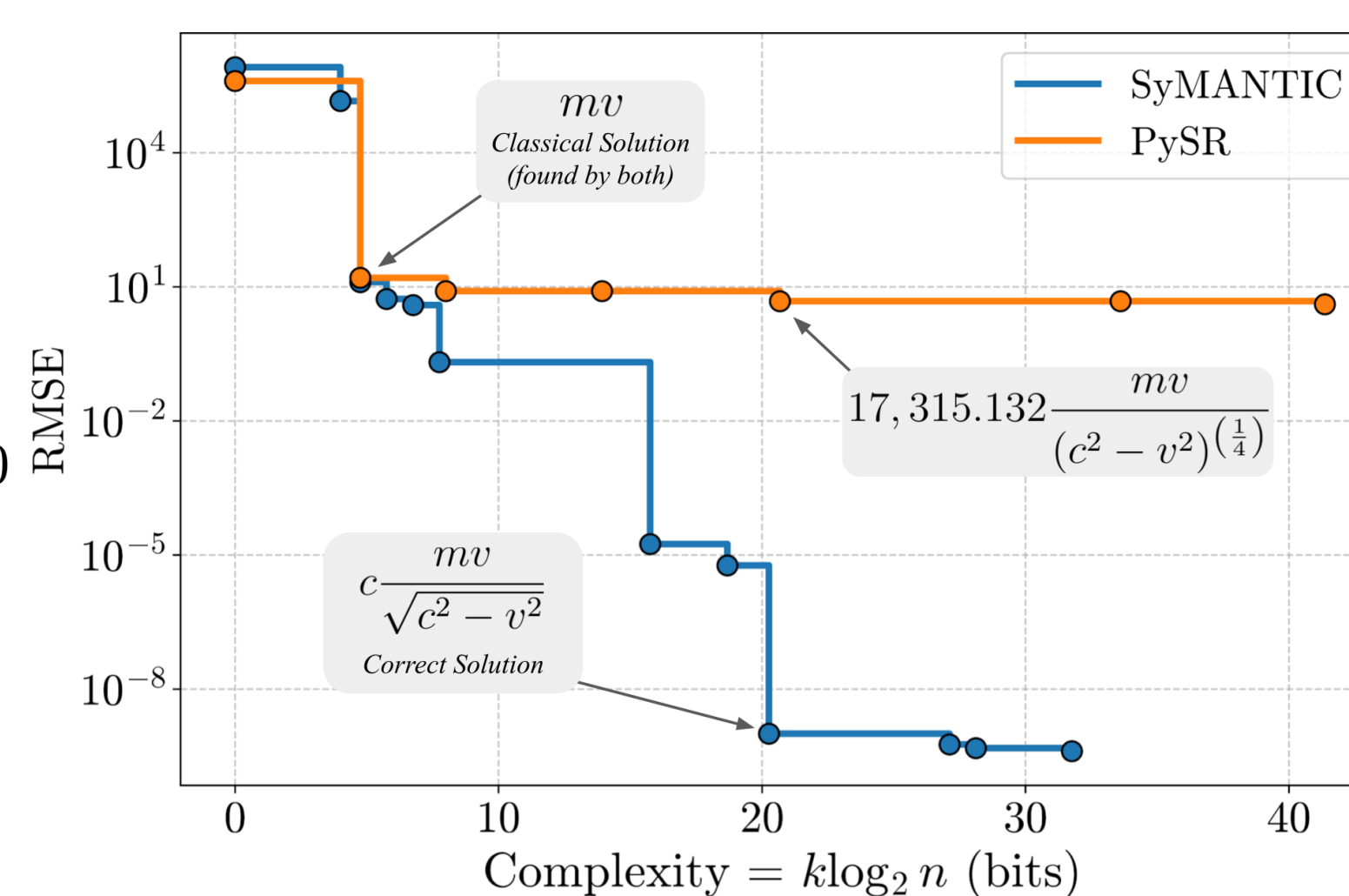
Equation Recovery

- We compare SyMANTIC with several state-of-the-art algorithms on numerous synthetic and physics equations.
- The main goal is to discover the ground truth equations.

Percent of equation recovery



Sparse data conditions



Classical and theoretical equation

Materials Exploration

Goal: Build an accurate, interpretable model that provides physical significance for driving predictions and generalization capabilities, which aids in the identification of new molecules.

